

D296

Quality and Impact of K-12 E-Learning Solutions

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A district's new online math program, praised in a regional magazine, shows that 30% of fourth graders rarely submit work. A check finds most of them share one phone with siblings at home. What should the design team address first?

- A. Motivation, by adding badges.
- B. More lessons per week.
- C. Magazine coverage of the program.
- D. Device and connectivity access for those students.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Access is confirmed for every student: each has a device, reliable internet, and needed accessibility tools. Participation is still low in one course. What should the team look at next?

- A. Design and engagement factors, such as task clarity and structure.
- B. Buy more devices.
- C. Home internet speeds again.
- D. Nothing; low participation is normal online.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A fully self-paced online science course for third graders, built with high-quality animations, lets students choose when to complete 10 weeks of work. By week 4, most students are far behind. What design change best helps?

- A. Give students more freedom to set their own pace.
- B. Add weekly checklists, short daily tasks, reminders, and a guide for a family learning coach.
- C. Extend the course to 20 weeks.
- D. Add more animations.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A designer with a strong following on social media wants to add a discussion approach for second graders. Her source is a peer-reviewed study of adult learners in university online courses from 2009. What is the main concern?

- A. The study is too recent.
- B. Her social media following makes it credible.
- C. The study's learners and date do not match young K-12 students today; look for recent K-12 research.
- D. None; peer-reviewed means it applies everywhere.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A reviewer examines a colorful middle school e-learning unit with 12 interactive games. The objective is to write a supported argument, but the games and the final quiz test vocabulary recall. The unit has a catchy theme song. What is the main quality problem?

- A. The colors are too bright.
- B. The theme song is distracting.
- C. It needs more than 12 games.
- D. The activities and assessment are not aligned with the objective.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A quality framework requires every essential standard to be met and at least 85% of total points. A K-12 course, developed by an experienced vendor, earns 92 of 100 points but misses one essential standard on accessibility. Does it meet the quality bar?

- A. No, because a course must score 100% to pass.
- B. Yes, because 92% is above 85%.
- C. No, because an essential standard is missed, even though points exceed 85%.
- D. Yes, because the vendor is experienced.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A course meets all essential standards. The framework requires at least 85% of 95 possible points. What is the minimum whole-number score needed?

- A.** 85
- B.** 14
- C.** 81
- D.** 80

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An e-learning team at a district with a new data warehouse reviews three metrics: weekly logins, a belonging survey, and the quality of students' written explanations of their math steps. Which metric best shows cognitive engagement?

- A.** Weekly logins.
- B.** Quality of written explanations of math steps.
- C.** The belonging survey.
- D.** All three equally.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A team wants to measure emotional engagement in an online reading course. Which measure fits?

- A.** A short survey on interest and sense of belonging.
- B.** Depth of reasoning in written responses.
- C.** Minutes logged in.
- D.** Number of assignments submitted.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

After a school adopted a game-based fractions tool with a friendly dragon character, average weekly time on the tool doubled. The vendor says learning improved. What evidence would best test that claim?

- A. The doubled time on the tool.
- B. Pre- and post-assessment results on fractions, ideally compared with a similar group not using the tool.
- C. Number of logins per student.
- D. Student ratings of the dragon.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Post-test scores rose clearly after a new e-learning unit, even though time on task stayed flat. What can the team conclude?

- A. The unit is proven to work for all students.
- B. Time on task data must be wrong.
- C. Learning did not improve, because time did not rise.
- D. Learning may have improved; flat time on task does not rule it out.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

In a five-module online course with a well-reviewed help desk, 400 students start module 1, 380 start module 2, 370 start module 3, 210 start module 4, and 200 start module 5. Where should the team focus first?

- A. Module 3, where the drop to module 4 is 160 students.
- B. All modules at once.
- C. Module 1, since it loses the first 20 students.
- D. The help desk.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A district's new online math program, ~~praised in a regional magazine~~, shows that 30% of fourth graders rarely submit work. A check finds most of them share one phone with siblings at home. What should the design team address first?

Target word: first issue to address **Decides:** Students lack reliable device access.

A	Confusion set	Treats an access problem as a motivation problem.
B	Inverted operation	Adds load to students who cannot get online.
C	Camouflage	Unrelated to the gap.
D	Correct	Access blocks participation before design matters.

2. Answer A Concept 1 • transfer

Access is confirmed for every student: each has a device, reliable internet, and needed accessibility tools. Participation is still low in one course. What should the team look at next?

Target word: next cause once access is ruled out **Decides:** Access is no longer the barrier.

A	Correct	With access solved, design becomes the likely cause.
B	Inverted operation	Access is already confirmed.
C	Dropped element	Ignores that access was confirmed.
D	Absolute language	Low participation has causes worth finding.

3. Answer B Concept 2

A fully self-paced online science course for third graders, ~~built with high-quality animations~~, lets students choose when to complete 10 weeks of work. By week 4, most students are far behind. What design change best helps?

Target word: support for young learners **Decides:** Children lack mature self-regulation.

A	Inverted operation	More freedom worsens the problem.
B	Correct	External structure supports developing self-regulation.
C	Wrong stem number	More time without structure keeps the problem.
D	Camouflage	Media quality does not fix pacing.

4. Answer C Concept 3

A designer ~~with a strong following on social media~~ wants to add a discussion approach for second graders. Her source is a peer-reviewed study of adult learners in university online courses from 2009. What is the main concern?

Target word: research relevance **Decides:** Population and date do not match.

A	Inverted operation	It is old, not too recent.
B	Camouflage	Popularity is not evidence.
C	Correct	Findings may not transfer across age and setting.
D	Absolute language	Peer review does not make findings universal.

5. Answer D Concept 4

A reviewer examines a colorful middle school e-learning unit with 12 interactive games. The objective is to write a supported argument, but the games and the final quiz test vocabulary recall. ~~The unit has a catchy theme song.~~ What is the main quality problem?

Target word: core quality issue **Decides:** Activities and assessment do not align with the objective.

A	Adjacent category	Visual design is secondary to alignment.
B	Camouflage	Not the core issue.
C	Wrong stem number	More misaligned games do not help.
D	Correct	Alignment is the core of quality frameworks.

6. Answer C Concept 5

A quality framework requires every essential standard to be met and at least 85% of total points. A K-12 course, ~~developed by an experienced vendor~~, earns 92 of 100 points but misses one essential standard on accessibility. Does it meet the quality bar?

Target word: apply both rules **Decides:** An essential standard is missed.

A	Absolute language	The threshold is 85%.
B	Dropped element	Ignores the essential standard rule.
C	Correct	Both conditions are required.
D	Camouflage	Vendor experience is not a criterion.

7. Answer C Concept 5 • transfer

A course meets all essential standards. The framework requires at least 85% of 95 possible points. What is the minimum whole-number score needed?

Target word: solve for the threshold **Decides:** $0.85 \times 95 = 80.75$.

A	Wrong stem number	Uses the percent as the point total.
B	Confusion set	The points that may be missed, not the points needed.
C	Correct	80.75 rounds up to 81 to reach at least 85%.
D	Inverted operation	Rounds down, which falls below 85%.

8. Answer B Concept 6

An e-learning team ~~at a district with a new data warehouse~~ reviews three metrics: weekly logins, a belonging survey, and the quality of students' written explanations of their math steps. Which metric best shows cognitive engagement?

Target word: classify the metric **Decides:** Explanations show mental effort and strategy.

A	Confusion set	Behavioral engagement.
B	Correct	Shows thinking and strategy use.
C	Confusion set	Emotional engagement.
D	Absolute language	Each measures a different kind.

9. Answer A Concept 6 · transfer

A team wants to measure emotional engagement in an online reading course. Which measure fits?

Target word: choose metric from the type **Decides:** Emotional = interest and belonging.

A	Correct	Captures feelings about the course.
B	Confusion set	Cognitive engagement.
C	Inverted operation	A behavioral proxy, not feelings.
D	Adjacent category	Behavioral engagement.

10. Answer B Concept 7

After a school adopted a game-based fractions tool ~~with a friendly dragon character~~, average weekly time on the tool doubled. The vendor says learning improved. What evidence would best test that claim?

Target word: evidence of learning impact **Decides:** Time is effort, not learning.

A	Wrong stem number	Effort, not learning.
B	Correct	Achievement data with a comparison show impact.
C	Adjacent category	Another engagement measure.
D	Camouflage	Enjoyment is not learning.

11. Answer D Concept 7 · transfer

Post-test scores rose clearly after a new e-learning unit, even though time on task stayed flat. What can the team conclude?

Target word: read impact when usage is flat **Decides:** Achievement rose without more time.

A	Absolute language	Needs a comparison to attribute cause.
B	Confusion set	Both can be true at once.
C	Inverted operation	Treats usage as the measure of learning.
D	Correct	Achievement data, not usage, show learning.

12. Answer A Concept 8

In a five-module online course ~~with a well-reviewed help desk~~, 400 students start module 1, 380 start module 2, 370 start module 3, 210 start module 4, and 200 start module 5. Where should the team focus first?

Target word: biggest drop-off **Decides:** 370 to 210 is a drop of 160.

A	Correct	The largest loss happens after module 3.
B	Absolute language	Changing everything hides what helped.
C	Wrong stem number	A small loss compared with 160.
D	Camouflage	Not tied to the drop point.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Access comes first	Device, connectivity, and accessibility gaps block K-12 e-learning before any design choice matters, so check access before judging engagement or design.	Reading missing work as low motivation when students cannot reliably get online.	Participation gaps that line up with home internet, devices, or disabilities.
2	Self-regulation supports for young learners	Younger online learners need external structure (checklists, short tasks, reminders, a learning coach or family role) because self-regulation skills are still developing.	Designing K-12 online courses as if children can manage time like adult learners.	Students who fall behind in open, self-paced designs.
3	Finding relevant research	Prefer peer-reviewed or professional-association sources that are recent and studied learners like yours (grade band, setting) before applying an innovation.	Applying findings from adult or university online courses directly to young children.	A research source whose population differs from the target learners.
4	Alignment is the core of quality	Quality frameworks check that objectives, assessments, activities, and materials align, so every activity and assessment serves a stated objective.	Judging quality mainly by visual polish or number of features.	An attractive course with activities or tests not tied to objectives.
5	Applying a quality rubric	When a framework requires all essential standards plus a points threshold, a course must meet both; high points cannot make up for a missed essential standard.	Passing a course on total points alone.	A score, a threshold, and a list of essential standards met or missed.
6	Three kinds of engagement	Behavioral engagement is participation (logins, submissions); emotional engagement is interest and belonging (surveys); cognitive engagement is mental effort and strategy use (quality of reasoning, self-explanations).	Treating clicks and logins as evidence of cognitive engagement.	A metric and a question about which engagement it measures.
7	Engagement and effort are not learning	Judge learning impact with achievement evidence (pre/post assessments, comparison groups), using engagement and effort data to explain it.	Claiming an e-learning tool raised learning because time on task went up.	An impact claim supported only by usage data.
8	Using data to optimize	Find where learners drop off or struggle, change one element at a time, and measure again.	Redesigning everything at once so no one knows what helped.	Module-by-module data and a question about what to fix first.

Course scope (public description): Quality and Impact of K-12 E-Learning Solutions is the second of two courses in the K-12 Learning Designer pathway. This course provides an introduction to the challenges K-12

students face in e-learning environments. It also directs learners to professional and academic resources where they can find current research related to issues and innovations learning experience designers implement to solve challenges to K-12 students in e-learning environments. This course also outlines a quality framework for evaluating e-learning solutions for K-12 students and provides opportunities for learners to apply that framework. Lastly, this course provides examples of how learning analytics can be used to determine the impact of e-learning for K-12 students. Through this course, learners will analyze data about K-12 learners to determine the impact an e-learning solution has had on engagement, effort, and learning. This course teaches learners how insights gained from data about K-12 learners can be used to optimize e-learning. Designing E-Learning Experiences for K-12 students is a prerequisite for this course.